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### INFORMATION PROCESSING APPARATUS, INFORMATION PROCESSING METHOD, AND INFORMATION PROCESSING PROGRAM

#### Abstract

An information processing apparatus in the present disclosure includes a processor, in which the processor is configured to: acquire a real-time video with an imaging apparatus; and control at least one of turning on or turning off light used to determine an irradiation range of radiation according to a movement of an object included in the acquired video.

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority under 35 USC 119 from Japanese Patent Application No. 2024-031605, filed on Mar. 1, 2024, the disclosure of which is incorporated by reference herein.

### BACKGROUND

#### 1. Technical Field

[0002] The present invention relates to an information processing apparatus, an information processing method, and an information processing program.

#### 2. Related Art

[0003] JP2014-064960A discloses an X-ray imaging apparatus comprising an X-ray tube that irradiates a subject with X-rays, a collimator that adjusts an X-ray emission field, a visible light source for visually recognizing the emission field with visible light, and a body thickness measuring device that measures a body thickness of the subject, in which the X-ray imaging apparatus further comprises a light source control device that turns off or turns on the visible light source in a case where a change in an output signal of the body thickness measuring device exceeds a determined threshold value.

[0004] JP2021-010541A discloses an X-ray diagnostic apparatus comprising an X-ray tube device that emits X-rays toward a subject, and an X-ray diaphragm device that comprises an X-ray diaphragm blade limiting an irradiation region of the X-rays applied to the subject, in which the X-ray diaphragm device comprises an irradiation field illumination unit that emits irradiation light for illuminating an irradiation range of the X-rays, an X-ray shielding box that accommodates the X-ray diaphragm blade and the irradiation field illumination unit and that has an opening serving as a passage for the X-rays and the irradiation light, and a light emitting unit that is disposed on an outer peripheral surface of the X-ray shielding box on an opening side.

[0005] WO2017/006535A discloses a radiography apparatus comprising a radiation irradiation device that irradiates a subject with radiation, imaging means that images the subject to acquire a captured image of the subject, a radiation detector that detects the radiation transmitted through the subject to generate a radiation image of the subject, and a driving situation controller that controls a driving situation of at least one of the radiation irradiation device or the radiation detector according to whether or not the radiation detector is present in the captured image. In addition, WO2017/006535A discloses the radiography apparatus that determines a part of the subject included in the captured image, and does not turn on an irradiation field lamp via a halfway push-operation of an imaging button in a case where the part is an animal face.

[0006] WO2020/115967A discloses an X-ray imaging apparatus comprising an X-ray irradiator that irradiates an imaging object with X-rays, light that irradiates the imaging object with visible light to indicate a range of the X-rays emitted from the X-ray irradiator, and a switch mechanism that is installed in a housing constituting the X-ray irradiator and that switches turning on and turning off the light, in which the switch mechanism comprises an insulating plate that is fixed to the housing in a state of penetrating an inside and an outside of the housing and that has a surface serving as the outside of the housing and a rear surface serving as the inside of the housing, an antenna that is installed on the rear surface of the insulating plate, and a controller that is connected between the antenna and the light and that controls at least one of turning on or turning off the light based on a change amount in capacitance of the antenna.

[0007] JP2011-245274A discloses an X-ray system comprising an imaging system including an X-ray radiation source, an X-ray image receptor, and a control circuit for controlling the X-ray radiation source, and a handheld interface device configured to wirelessly communicate with the imaging system, in which the handheld interface device is configured to receive a user input

command for operating the imaging system, and the imaging system is configured to wirelessly receive the command and execute the command to operate the imaging system.

## SUMMARY

[0008] In radiography such as X-raying, a user such as a radiologist who operates a radiography apparatus turns on light provided in the radiography apparatus to determine a range in which a subject is irradiated with radiation. In a case where the irradiation range is determined, the user can acquire a radiation image of the subject by turning off the light and irradiating the subject with the radiation.

[0009] An object of the present disclosure is to provide an information processing apparatus that can easily control turning on and turning off light remotely, as compared with a case where the light in a case of determining a range in which radiation is emitted is turned on and turned off by a switch.

[0010] In order to achieve the above object, an information processing apparatus according to a first aspect comprises a processor, in which the processor is configured to: acquire a real-time video with an imaging apparatus; and control at least one of turning on or turning off light used to determine an irradiation range of radiation according to a movement of an object included in the acquired video.

[0011] In an information processing apparatus according to a second aspect, in the information processing apparatus according to the first aspect, in a case where a presence of a person is detected as the movement of the object from the acquired video, the processor controls the light to be turned on.

[0012] In an information processing apparatus according to a third aspect, in the information processing apparatus according to the first aspect, in a case where a movement of a person is detected as the movement of the object from the acquired video, the processor controls the light to be turned on.

[0013] In an information processing apparatus according to a fourth aspect, in the information processing apparatus according to the first aspect, in a case where a specific movement of a person is detected as the movement of the object from the acquired video, the processor controls the light to be turned on or turned off.

[0014] In an information processing apparatus according to a fifth aspect, in the information processing apparatus according to the first aspect, in a case where a movement of a plurality of persons is detected as the movement of the object from the acquired video, the processor recognizes a specific movement of a specific person from among the plurality of persons, and controls the light to be turned on or turned off based on the recognized specific movement of the specific person.

[0015] In an information processing apparatus according to a sixth aspect, in the information processing apparatus according to the first aspect, the processor is configured to: limit a range in which the video is acquired to a size of an image receiving unit for the radiation or to the irradiation range of the radiation; and control turning on and turning off the light based on the video acquired in the limited range.

[0016] In an information processing apparatus according to a seventh aspect, in the information processing apparatus according to any one of the first to sixth aspects, the processor is configured to control turning on and turning off the light based on a first mode in which the light is turned on and turned off using the acquired video or on a second mode in which the light is not turned on and turned off using the acquired video.

[0017] In an information processing apparatus according to an eighth aspect, in the information processing apparatus according to the seventh aspect, the processor is configured to control turning on and turning off the light via the first mode in a state in which radiography is possible.

[0018] In an information processing apparatus according to a ninth aspect, in the information processing apparatus according to the seventh aspect, the processor is configured to control turning

on and turning off the light via the first mode in a case of radiography in a specific imaging menu.

[0019] In an information processing apparatus according to a tenth aspect, in the information processing apparatus according to the seventh aspect, the processor is configured to control turning on and turning off the light via the first mode in a case of radiography at an upright position.

[0020] In an information processing apparatus according to an eleventh aspect, in the information processing apparatus according to the seventh aspect, the video acquired by the imaging apparatus is a visible light image.

[0021] In an information processing apparatus according to a twelfth aspect, in the information processing apparatus according to the seventh aspect, the video acquired by the imaging apparatus is a visible light image and a depth image corresponding to the visible light image.

[0022] An information processing program according to a thirteenth aspect causes a computer to execute a process comprising: acquiring a real-time video with an imaging apparatus; and controlling at least one of turning on or turning off light used to determine an irradiation range of radiation according to a movement of an object included in the acquired video.

[0023] An information processing method according to a fourteenth aspect is executed by a computer, the method comprising: a step of acquiring a real-time video with an imaging apparatus; and a step of controlling at least one of turning on or turning off light used to determine an irradiation range of radiation according to a movement of an object included in the acquired video.

[0024] With the information processing apparatus of the first aspect, an effect is exhibited that a user can easily control turning on and turning off light remotely, as compared with a case where the user turns on and turns off the light in a case of determining a range in which radiation is emitted by a switch.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0025] Exemplary embodiments of the technology of the disclosure will be described in detail based on the following figures, wherein:

[0026] FIG. 1 is a diagram showing an outline of a radiography system according to the present embodiment.

[0027] FIG. 2 is a diagram showing a schematic configuration of an imaging apparatus according to the present embodiment.

[0028] FIG. 3 is a diagram showing a hardware configuration of a controller and a console according to the present embodiment.

[0029] FIG. 4 is a flowchart showing an example of a process executed by an information processing program.

[0030] FIG. 5 is a flowchart showing an example of a control process of a light irradiation unit via a first mode; and

[0031] FIG. 6 is a flowchart showing an example of the control process of the light irradiation unit via a second mode.

### DETAILED DESCRIPTION

[0032] Hereinafter, embodiments of the present disclosure will be described in detail with reference to the accompanying drawings.

[0033] FIG. 1 is a diagram showing a schematic configuration of a radiography system 1. As shown in FIG. 1, the radiography system 1 comprises an imaging apparatus 10 and a console 30. The imaging apparatus 10 and the console 30, and the console 30 and an external radiology information system (RIS) 50 are communicatively connected to each other via a wired or wireless network.

[0034] The console 30 acquires the imaging order or the like from the RIS 50. In addition, the

console **30** controls the imaging apparatus **10** according to the acquired imaging order, the setting and the instruction designated by the user, and the like, and performs the radiography process. In a case where the radiography process is started, the imaging apparatus **10** performs the radiography in which a subject H is irradiated with radiation under the control of the console **30** and a radiation image of the subject H is acquired. The console **30** is an example of an “information processing apparatus”.

[0035] The console **30** is operated by a person who handles the imaging apparatus **10**, for example, a radiologist. The radiologist is an example of a “user”. The radiography system **1** may include picture archiving and communication systems (PACS) (not shown).

[0036] Next, the imaging apparatus **10** will be explained with reference to FIG. 2. FIG. 2 is a diagram showing a schematic configuration of the imaging apparatus **10**. FIG. 2 shows an example of a state in which the chest of the patient indicating the subject H is radiographed as the imaging part.

[0037] As shown in FIG. 2, the imaging apparatus **10** comprises a radiation emitting unit **11** and a radiation detector **15**. The radiation emitting unit **11** includes a radiation source **12** that emits radiation such as X-rays, and an optical camera **13**. The optical camera **13** is an example of an “imaging apparatus”. In addition, the radiation detector **15** is an example of an “image receiving unit”.

[0038] In a case where the user gives an instruction to emit the radiation, the imaging apparatus **10** emits the radiation from the radiation source **12**. A range R shown in FIG. 2 indicates a region where radiation is emitted. The range R is adjusted to be changeable by the user. The type of the radiation source **12** is not particularly limited, and for example, a radiation source such as a hot cathode type or a cold cathode type can be appropriately applied.

[0039] The optical camera **13** is an optical digital camera that includes, for example, a complementary metal oxide semiconductor (CMOS) image sensor or a charge coupled device (CCD) image sensor and that performs imaging based on visible light. As the optical camera **13**, a 2D camera such as a Color camera, a 3D camera comprising a Color camera and a Depth camera, or the like can be appropriately applied. The optical camera **13** enables the still image capture and the motion picture capture. The optical camera **13** can image the periphery of the radiation emitting unit **11** and acquire a visible light image, a video of a depth image corresponding to the visible light image, and the like in real time.

[0040] In the radiography system **1**, the movement of a person, an object, and the like included in the image and the video acquired by the optical camera **13** can be detected using, for example, an inter-frame difference, optical flow, a posture estimation system, and the like.

[0041] In addition, in the radiography system **1**, the presence of the person, the object, and the like included in the image and the video acquired by the optical camera **13** can be detected using, for example, a human body detection system, a posture estimation system, and the like.

[0042] The optical camera **13** can image a region wider than the range R shown in FIG. 2, for example, a range C shown in FIG. 2.

[0043] The position at which the optical camera **13** is installed is not particularly limited. For example, the optical camera **13** shown in FIG. 2 may be attached to the same surface as the irradiation opening for radiation of the radiation emitting unit **11**. Alternatively, the optical camera **13** may be attached to a wall surface or the like of the imaging room.

[0044] The radiation detector **15** detects the radiation transmitted through the subject H on a detection surface **15A**. The radiation detector **15** generates the radiation image based on the detected radiation, and outputs the generated radiation image to, for example, the radiation emitting unit **11** and the console **30**.

[0045] The radiation detector **15** is, for example, a portable electronic cassette, and may be used by being placed on any seat or being held by the subject H. In addition, the radiation detector **15** may be movable to any position in the horizontal direction (the X direction and the Y direction) and the

vertical direction (the Z direction) with respect to the radiation emitting unit **11**. In addition, the radiation detector **15** may be a stationary type to be disposed inside an imaging table installed in the imaging room.

[0046] The type of the radiation detector **15** is not particularly limited. The radiation detector **15** may be an indirect-conversion-type radiation detector that converts the radiation into light and converts the converted light into a charge, or may be a direct-conversion-type radiation detector that directly converts the radiation into a charge.

[0047] The radiation emitting unit **11** may be a ceiling-running type emitting unit that is held by a support column suspended from a ceiling of an imaging room. In the ceiling-running type emitting unit, a support column that is expandable and contractible in the vertical direction (Z direction) is attached to rails running around the ceiling via wheels, and the emitting unit is movable in the horizontal direction (X direction and Y direction) in the imaging room. By the movement of the support column in the horizontal direction and the expansion and contraction of the support column in the vertical direction, the radiation emitting unit **11** is also translationally moved in the horizontal direction and the vertical direction. In addition, the radiation emitting unit **11** may be movable rotationally around a rotational axis extending in the horizontal direction, or may be movable rotationally around a rotational axis extending in the vertical direction.

[0048] In addition, the radiation emitting unit **11** may be, for example, a portable emitting unit. The portable emitting unit may be used for, for example, a simple radiographic examination in a medical facility, a radiographic examination in a case of home medical care, a radiographic examination outdoors, an on-site medical care in a disaster-stricken area or a medically underserved area, or the like. In addition, for example, the radiation emitting unit **11** may be a stationary emitting unit installed in an imaging room. The operation of the radiation emitting unit **11** is controlled by the console **30**.

[0049] FIG. **3** is a diagram showing an example of a hardware configuration of the console **30** that controls the operation of the radiation emitting unit **11** and the operation of the imaging apparatus **10**. For description, FIG. **3** also shows a part of a hardware configuration of the imaging apparatus **10**.

[0050] The console **30** includes a central processing unit (CPU) **31**, a memory **32** such as a read only memory (ROM) or a random access memory (RAM), a storage unit **33**, a display **34**, an operation unit **35**, an interface (I/F) unit **36**, and the like.

[0051] The CPU **31**, for example, expands various programs including an information processing program **100** stored in the storage unit **33** to the memory **32** and controls each configuration of the radiation emitting unit **11**.

[0052] The display **34** displays the image and the video captured by the optical camera **13** in real time. The display **34** may display the radiation image generated in the radiation detector **15**.

[0053] The I/F unit **36** is an interface for connecting each configuration provided in the imaging apparatus **10** and the console **30** to each other in a communicable manner. In addition, the I/F unit **36** is an interface for connecting the console **30** and the RIS **50** in a communicable manner.

[0054] The CPU **31** can control turning on and turning off a light irradiation unit **14** provided in the radiation emitting unit **11**. The user can turn on the light irradiation unit **14** to determine the position of the range R that is the irradiation region of the radiation, and the like.

[0055] The light irradiation unit **14** is provided with, for example, a timer (not shown). After the light irradiation unit **14** is started to be turned on, the timer starts counting, and after a predetermined time such as **30** seconds elapses, the light irradiation unit **14** is automatically turned off. The light irradiation unit **14** is an example of “light”. The light irradiation unit **14** is composed of light emitting diode (LED) light, laser diode (LD) light, and the like.

[0056] The operation unit **35** is an interface between each component provided in the console **30** and the imaging apparatus **10** and the user. The turning on and turning off of the light irradiation unit **14** can be controlled by, for example, a switch provided in the console **30**, the imaging

apparatus **10**, and the like.

[0057] In the console **30**, the information on the patient can be acquired from the RIS **50**, and the acquired information on the patient can be displayed on the display **34**. The user can input information on a patient or the like serving as the subject H in the console **30** and can acquire the information on the patient from the RIS **50**. In addition, the CPU **31** is an example of a “processor”. [0058] Next, a process of the information processing program **100** executed by the CPU **31** will be described with reference to FIG. **4**.

[0059] FIG. **4** is a flowchart showing an example of a process executed by the CPU **31** with the information processing program **100**.

[0060] In step **S100**, the CPU **31** receives the setting information from the user. Here, the setting information includes various settings in the imaging apparatus **10** in a case of imaging the subject H, such as an imaging menu and a light control mode for controlling turning on and turning off the light irradiation unit **14**. The setting information is registered by the user. In addition, the setting information includes information on a patient serving as the subject H. The setting information is registered in, for example, the console **30**.

[0061] In the imaging menu, a part of the patient serving as the subject H to be imaged, such as the head, the chest, the abdomen, and the leg, can be registered. In the light control mode, it is possible to select and register whether or not to use the first mode as the light control mode for controlling turning on and turning off the light irradiation unit **14**. According to the light control in the first mode, it is possible to control turning on and turning off the light using the video acquired from the optical camera **13**. Details of the control of the light irradiation unit **14** in the first mode will be described later (see FIG. **5**). The user can control turning on and turning off the light irradiation unit **14** by operating the operation unit **35** regardless of whether or not the first mode is set. A mode in which the turning on and turning off of the light irradiation unit **14** are controlled by a mode other than the first mode, in other words, the turning on and turning off of the light irradiation unit **14** are controlled by the operation of the operation unit **35** without using the video, will be hereinafter referred to as a second mode. Details of the control of turning on and turning off the light irradiation unit **14** via the second mode will be described later (see FIG. **6**).

[0062] In step **S102**, the CPU **31** receives an instruction to start the examination. The instruction to start the examination is given, for example, by the user operating the console **30**. In a case where the instruction to start the examination is received, the CPU **31** starts the radiography process with the radiography system **1**.

[0063] In step **S104**, the CPU **31** determines whether or not the first mode is set as the control mode of the light irradiation unit **14** in the setting information. In a case where the first mode is set, the CPU **31** affirms the determination. In a case where the determination is affirmed, the CPU **31** executes the process of step **S106**. In a case where the first mode is not set, the CPU **31** denies the determination, and executes the process of step **S108**.

[0064] In step **S106**, the CPU **31** changes the control mode of the light irradiation unit **14** to the first mode.

[0065] In step **S108**, the CPU **31** determines whether or not the radiation has been emitted. The irradiation instruction of the radiation is given by the user. In a case where the radiation is emitted, the CPU **31** affirms the determination of step **S108**. In a case where the emission of the radiation has not been completed, the CPU **31** denies the determination of step **S108**.

[0066] In a case where the determination of step **S108** is affirmed, the CPU **31** executes the process of step **S110**. In a case where the determination of step **S108** is denied, the CPU **31** executes the process of step **S108**.

[0067] In step **S110**, the CPU **31** acquires the radiation image generated in the radiation detector **15**.

[0068] In step **S112**, the CPU **31** determines whether or not the control mode of the light irradiation unit **14** is set to the first mode. The case where the control mode of the light irradiation unit **14** is

set to the first mode is, for example, a case where the determination is affirmed in the process of step **S104**. In a case where the control mode of the light irradiation unit **14** is set to the first mode, the CPU **31** affirms the determination of step **S112** and executes the process of step **S114**. In a case where the control mode of the light irradiation unit **14** is not set to the first mode, the CPU **31** denies the determination of the process of step **S112** and executes the process of step **S116**.

[0069] In step **S114**, the CPU **31** ends the control of the light irradiation unit **14** via the first mode.

[0070] In step **S116**, the CPU **31** receives an instruction to end the examination, for example, via an instruction from the user. After the instruction to end the examination is received, the CPU **31** ends the radiography process with the radiography system **1**. The CPU **31** executes the process shown in FIG. **4** each time the radiography is performed.

[0071] With the above process, in a case where the control mode of the light irradiation unit **14** is set to the first mode, the CPU **31** can control the light irradiation unit **14** in the first mode, for example, while executing the process from step **S106** to step **S110**. Under the control of the light irradiation unit **14** via the first mode, the user can determine a range in which the radiation is emitted and can acquire the radiation image of the subject H.

[0072] Next, a detailed description of process of controlling the light irradiation unit **14** via the first mode will be made with reference to FIG. **5**. The process shown in FIG. **5** is executed by the CPU **31**, for example, in a case where the determination of step **S104** of FIG. **4** is affirmed.

[0073] In step **S200**, the CPU **31** acquires the video captured by the optical camera **13** in real time.

[0074] In step **S202**, the CPU **31** determines whether or not a turn-on instruction indicating an instruction to turn on the light irradiation unit **14** has been received by the operation of the operation unit **35** of the user. In a case where the turn-on instruction is received by the operation of the operation unit **35** of the user, the CPU **31** affirms the determination and executes the process of step **S206**. In a case where the operation unit **35** is not operated and the turn-on instruction is not received by the operation unit **35**, the CPU **31** denies the determination and executes the process of step **S204**.

[0075] In the process of step **S204**, the CPU **31** determines whether or not at least one of the presence or movement of the object or the person included in the acquired video is detected.

[0076] Specifically, for example, in a case where the presence of the object or the person is detected in the acquired video, the CPU **31** affirms the determination. In addition, as another example, in a case where a specific person is present in the acquired video, the CPU **31** affirms the determination. An example of the specific person is a patient serving as the subject H or a user who operates the imaging apparatus **10** or the like in a case of imaging an affected part, such as a radiologist. In addition, for example, in a case where the movement of any person and any object included in the acquired video is detected, the CPU **31** affirms the determination. In addition, as another example, for example, in a case where persons such as a patient serving as the subject H and a user included in the acquired video are recognized, and a movement of a predetermined specific person among the recognized persons is detected, the CPU **31** affirms the determination. Here, the subject H, the patient serving as the subject H, and the user are examples of “objects”. In addition, the patient serving as the subject H and the user are examples of “persons”.

[0077] In addition, as another example, in a case where a gesture, which indicates a turn-on instruction, of any person included in the video, a recognized predetermined specific person, or the like is detected, the CPU **31** affirms the determination. The gesture indicating the turn-on instruction is an example of a “specific movement”. The association between the turn-on instruction of the light irradiation unit **14** and the gesture is set in advance by the user and is stored in, for example, the storage unit **33**.

[0078] In the process of step **S204**, for example, the presence and movement of an object, a person, or the like included in the entire region of the range C imaged by the optical camera **13** may be a target of detection. Alternatively, the presence and movement of an object, a person, or the like included in a limited range such as a part of the range C, for example, a range corresponding to the



radiation detector **15**, the range R serving as the irradiation region of radiation, and the like, may be a target of detection. With the above configuration, it is possible to prevent the light from being unnecessarily turned on and turned off, compared to a case where the range in which the video is acquired is not limited.

[0079] In a case where the movement is detected in the process of step **S204**, the CPU **31** affirms the determination and executes the process of step **S206**. In a case where the movement is not detected in the process of step **S204**, the CPU **31** denies the determination and executes the process of step **S202**.

[0080] In step **S206**, the CPU **31** turns on the light irradiation unit **14**.

[0081] In the process of step **S208**, the CPU **31** determines whether or not a turn-off instruction indicating an instruction to turn off the light irradiation unit **14** has been received by pressing a switch provided in the operation unit **35** or the like. In a case where the turn-off instruction is received, the CPU **31** affirms the determination and executes the process of step **S218**. In a case where the turn-off instruction is not received, the CPU **31** denies the determination and executes the process of step **S210**.

[0082] In the process of step **S210**, the CPU **31** determines whether or not the movement of the person or the object included in the acquired video is detected. The details of the movement of the person or the object included in the video are the same as the description of step **S204**, and thus the description thereof will be omitted.

[0083] In a case where the movement is detected in the process of step **S210**, the CPU **31** affirms the determination and executes the process of step **S212**. On the other hand, in a case where the movement is not detected, the CPU **31** denies the determination and executes the process of step **S216**.

[0084] In the process of step **S212**, the CPU **31** determines whether or not the turn-off instruction is received from the detected movement. For example, in a case where a gesture is detected as the movement in the process of step **S210**, in a case where the gesture indicates the turn-off instruction, the CPU **31** affirms the determination. Meanwhile, in a case where the movement of the person included in the video is detected, in a case where the movement is not a movement indicating the turn-off instruction, the determination is denied. The association between the turn-off instruction of the light irradiation unit **14** and the gesture is set in advance by the user and is stored in, for example, the storage unit **33**.

[0085] In a case where the determination of the process of step **S212** is affirmed, the CPU **31** executes the process of step **S218**. In a case where the determination of the process of step **S212** is denied, the CPU **31** executes the process of step **S214**.

[0086] In the process of step **S212**, for example, the presence and movement of the object, the person, or the like included in the entire region of the range C imaged by the optical camera **13** may be a target of detection. Alternatively, the presence and movement of an object, a person, or the like included in a limited range such as a part of the range C, for example, a range corresponding to the radiation detector **15**, the range R serving as the irradiation region of radiation, and the like, may be a target of detection. With the above configuration, it is possible to prevent the light from being unnecessarily turned on and turned off, compared to a case where the range in which the video is acquired is not limited.

[0087] In step **S214**, the CPU **31** resets the count of the timer of the light irradiation unit **14**.

[0088] In step **S216**, the CPU **31** determines whether or not a predetermined time has elapsed since the start of turning on the light irradiation unit **14**. An example of the predetermined time is a predetermined time, for example, **30** seconds. In a case where the predetermined time has elapsed, the CPU **31** affirms the determination and executes the process of step **S218**. In a case where the predetermined time has not elapsed, the CPU **31** denies the determination and executes the process of step **S208**.

[0089] In step **S218**, the CPU **31** turns off the light irradiation unit **14**. While the process of FIG. **4**

is being executed, the CPU **31** repeatedly executes the process shown in FIG. **5** while the control via the first mode is being continued.

[0090] As described above, according to the process of step **S200** to step **S218**, the CPU **31** can control turning on and turning off the light irradiation unit **14** using the video acquired by the optical camera **13** in addition to the control turning on and turning off the light irradiation unit **14** via the user pressing the switch.

[0091] In the process of step **S204**, step **S210**, and step **S212**, in a case where the video includes a plurality of persons, the CPU **31** may be configured to detect a movement limited to a specific person such as the user and perform the determination. With the above configuration, in a case where the plurality of persons are included in the video, it is possible to prevent the light from being turned on or turned off by the movement other than the movement of the specific person.

[0092] Next, a detailed description of process of controlling the light irradiation unit **14** via the process of the second mode will be made with reference to FIG. **6**. The process shown in FIG. **6** is executed by the CPU **31**, for example, while the process of FIG. **4** is being executed. The process shown in FIG. **6** is not executed while the control via the first mode is continued in a case where the determination of step **S104** is affirmed.

[0093] In step **S300**, the CPU **31** determines whether or not the turn-on instruction of the light irradiation unit **14** is received from the operation unit **35**. In a case where an instruction to turn on the light irradiation unit **14** is given by pressing a button provided in the operation unit **35**, or the like, the CPU **31** affirms the determination.

[0094] In a case where the CPU **31** affirms the determination, the CPU **31** executes the process of step **S302**. In a case where the CPU **31** denies the determination, the CPU **31** executes the process of step **S300**.

[0095] In step **S302**, the CPU **31** turns on the light irradiation unit **14**.

[0096] In step **S304**, the CPU **31** determines whether or not the turn-off instruction of the light irradiation unit **14** is received. In a case where the turn-off instruction of the light irradiation unit **14** is received by pressing a button provided in the operation unit **35**, or the like, the CPU **31** affirms the determination.

[0097] In a case where the CPU **31** affirms the determination, the CPU **31** executes the process of step **S308**. In a case where the CPU **31** denies the determination, the CPU **31** executes the process of step **S306**.

[0098] In step **S306**, the CPU **31** determines whether or not a predetermined time has elapsed since the start of turning on the light irradiation unit **14**. The predetermined time is, for example, **30** seconds. In a case where the predetermined time has elapsed since the start of turning on the light irradiation unit **14**, the CPU **31** affirms the determination of step **S306**.

[0099] In a case where the CPU **31** affirms the determination, the CPU **31** executes the process of step **S308**. In a case where the CPU **31** denies the determination, the CPU **31** executes the process of step **S304**.

[0100] In step **S308**, the CPU **31** controls the light irradiation unit **14** to be turned off. While the process of FIG. **4** is being executed, the process shown in FIG. **6** is repeatedly executed.

[0101] As described above, according to the process of steps **S300** to **S308**, the CPU **31** can control turning on and turning off the light irradiation unit **14** by, for example, the operation of the operation unit **35** of the user. With the second mode, the CPU **31** does not control turning on and turning off the light irradiation unit **14** using the video acquired by the optical camera **13**.

[0102] As described above, according to the process shown in FIGS. **4** to **6**, at least one of the turning on or turning off of the light irradiation unit **14** can be controlled using the video acquired by the optical camera **13** according to the user's settings. In addition, with the imaging apparatus **10** equipped with the optical camera **13**, the user can control turning on and turning off the light irradiation unit **14** remotely without adding a dedicated input device such as an ultrasonic distance sensor, an optical distance sensor, and a voice input microphone.

[0103] In addition, according to the control of the light irradiation unit **14** based on the video (the control of the light irradiation unit **14** via the first mode), it is possible to automatically turn on the light according to the presence of a person, as compared with a case where the user turns on the light with the switch. In addition, according to the control of the light irradiation unit **14** based on the video (the control of the light irradiation unit **14** via the first mode), in a case where the light used in a case of determining the range in which the radiation is emitted is automatically turned off by the timer, the user can easily turn on the light as compared with a case where the user turns on the light again by using the switch.

[0104] In addition, according to the control of the light irradiation unit **14** based on the video (the control of the light irradiation unit **14** via the first mode), the user can easily control turning on and turning off the light remotely, as compared with a case where the user turns on the light with a switch and automatically turns off the light with a timer. In addition, according to the control of the light irradiation unit **14** based on the video (the control of the light irradiation unit **14** via the first mode), in a case where the plurality of persons are included in the video, it is possible to prevent the light from being turned on or turned off by the movement other than the movement of the specific person.

[0105] In addition, according to the control of the light irradiation unit **14** based on the video (the control of the light irradiation unit **14** via the first mode), it is possible to prevent the light from being unnecessarily turned on and turned off, compared to a case where the range in which the video is acquired is not limited.

[0106] In addition, with the configuration in which the first mode can be set as the control mode of the light irradiation unit **14** in the setting information, the user can select whether to control turning on and turning off the light using the acquired video or to control turning on and turning off the light without using the acquired video.

[0107] In addition to the above, a configuration may be adopted in which the timing of controlling turning on and turning off the light irradiation unit **14** in the first mode is appropriately changed by, for example, the user's settings.

[0108] For example, in a case where the imaging apparatus **10** is available for performing imaging, in other words, the imaging apparatus **10** is available for performing radiography, the CPU **31** may be configured to control turning on and turning off the light irradiation unit **14** in the first mode. The case in which the imaging apparatus **10** is available for performing imaging indicates, for example, a period from the registration of the setting information via the user in the console **30** to the start of the emission of the radiation. More specifically, the case indicates a period in which the process from step **S102** to step **S108** in FIG. **4** is executed. With the above-described configuration, it is possible to prevent the light from being unnecessarily turned on, compared to a case where the turning on and turning off of the light are controlled by the first mode at all times.

[0109] In addition, for example, in a case where a specific imaging menu is set in the imaging menu included in the setting information, the CPU **31** may be configured to control turning on and turning off the light irradiation unit **14** in the first mode. An example of the specific imaging menu is chest imaging, abdominal imaging, and the like. With the above-described configuration, it is possible to prevent the light from being unnecessarily turned on, compared to a case where the turning on and turning off of the light are controlled by the first mode at all times.

[0110] In addition, for example, in a case where the imaging method can be set in the imaging menu included in the setting information and the imaging method is a specific imaging method, the CPU **31** may be configured to control turning on and turning off the light irradiation unit **14** in the first mode. Examples of the imaging method include upright imaging without using a bed and decubitus imaging using a bed. For example, the user can set the light irradiation unit **14** to be controlled to be turned on and turned off in the first mode in a case of upright imaging. With the above-described configuration, it is possible to prevent the light from being unnecessarily turned on, compared to a case where the turning on and turning off of the light are controlled by the first

mode at all times.

[0111] According to the present embodiment, it is possible to control turning on and turning off the light by using at least one of the video corresponding to the visible light image or the video of the depth image corresponding to the visible light image.

[0112] In any case, the light irradiation unit **14** is turned off during the emission of the radiation.

[0113] The process shown in FIGS. **4**, **5**, and **6** has been described as being executed by the CPU **31** of the console **30**, but the process with the information processing program **100** may be executed by the imaging apparatus **10**. In a case where the imaging apparatus **10** executes the process, the imaging apparatus **10** comprises a controller capable of executing the information processing program **100**.

[0114] In the present embodiment, the information processing program **100** has been described as being stored in the storage unit **33**, but the information processing program **100** may be provided by a storage medium such as a CD-ROM, or may be downloaded via a network.

#### Supplementary Note

##### [0115] Supplementary Note 1

[0116] An information processing apparatus comprising: [0117] a processor, [0118] in which the processor is configured to: [0119] acquire a real-time video with an imaging apparatus; and [0120] control at least one of turning on or turning off light used to determine an irradiation range of radiation according to a movement of an object included in the acquired video.

##### Supplementary Note 2

[0121] The information processing apparatus according to Supplementary Note 1, [0122] in which, in a case where a presence of a person is detected as the movement of the object from the acquired video, the processor controls the light to be turned on.

##### Supplementary Note 3

[0123] The information processing apparatus according to Supplementary Note 1, [0124] in which, in a case where a movement of a person is detected as the movement of the object from the acquired video, the processor controls the light to be turned on.

##### Supplementary Note 4

[0125] The information processing apparatus according to Supplementary Note 1, [0126] in which, in a case where a specific movement of a person is detected as the movement of the object from the acquired video, the processor controls the light to be turned on or turned off.

##### Supplementary Note 5

[0127] The information processing apparatus according to Supplementary Note 1, [0128] in which, in a case where a movement of a plurality of persons is detected as the movement of the object from the acquired video, the processor recognizes a specific movement of a specific person from among the plurality of persons, and [0129] controls the light to be turned on or turned off based on the recognized specific movement of the specific person.

##### Supplementary Note 6

[0130] The information processing apparatus according to any one of Supplementary Note 1 to Supplementary Note 5, [0131] in which the processor is configured to: [0132] limit a range in which the video is acquired to a size of an image receiving unit for the radiation or to the irradiation range of the radiation; and [0133] control turning on and turning off the light based on the video acquired in the limited range.

##### Supplementary Note 7

[0134] The information processing apparatus according to any one of Supplementary Note 1 to Supplementary Note 6, [0135] in which the processor is configured to control turning on and turning off the light based on a first mode in which the light is turned on and turned off using the acquired video or on a second mode in which the light is not turned on and turned off using the acquired video.

##### Supplementary Note 8

[0136] The information processing apparatus according to Supplementary Note 7, [0137] in which the processor is configured to control turning on and turning off the light via the first mode in a state in which radiography is possible.

#### Supplementary Note 9

[0138] The information processing apparatus according to Supplementary Note 7, [0139] in which the processor is configured to control turning on and turning off the light via the first mode in a case of radiography in a specific imaging menu.

#### Supplementary Note 10

[0140] The information processing apparatus according to Supplementary Note 7, [0141] in which the processor is configured to control turning on and turning off the light via the first mode in a case of radiography at an upright position.

#### Supplementary Note 11

[0142] The information processing apparatus according to any one of Supplementary Note 1 to Supplementary Note 10, [0143] in which the video acquired by the imaging apparatus is a visible light image.

#### Supplementary Note 12

[0144] The information processing apparatus according to any one of Supplementary Note 1 to Supplementary Note 10, [0145] in which the video acquired by the imaging apparatus is a visible light image and a depth image corresponding to the visible light image.

#### Supplementary Note 13

[0146] An information processing program causing a computer to execute a process comprising:

[0147] acquiring a real-time video with an imaging apparatus; and [0148] controlling at least one of turning on or turning off light used to determine an irradiation range of radiation according to a movement of an object included in the acquired video.

#### [0149] Supplementary Note 14

[0150] An information processing method executed by a computer, the method comprising: [0151] a step of acquiring a real-time video with an imaging apparatus; and [0152] a step of controlling at least one of turning on or turning off light used to determine an irradiation range of radiation according to a movement of an object included in the acquired video.

## Claims

1. An information processing apparatus comprising: a processor, wherein the processor is configured to: acquire a real-time video with an imaging apparatus; and control at least one of turning on or turning off light used to determine an irradiation range of radiation according to a movement of an object included in the acquired video.
2. The information processing apparatus according to claim 1, wherein, in a case where a presence of a person is detected as the movement of the object from the acquired video, the processor controls the light to be turned on.
3. The information processing apparatus according to claim 1, wherein, in a case where a movement of a person is detected as the movement of the object from the acquired video, the processor controls the light to be turned on.
4. The information processing apparatus according to claim 1, wherein, in a case where a specific movement of a person is detected as the movement of the object from the acquired video, the processor controls the light to be at least one of turned on or turned off.
5. The information processing apparatus according to claim 1, wherein, in a case where a movement of a plurality of persons is detected as the movement of the object from the acquired video, the processor recognizes a specific movement of a specific person from among the plurality of persons, and controls the light to be at least one of turned on or turned off based on the recognized specific movement of the specific person.

- 6.** The information processing apparatus according to claim 1, wherein the processor is configured to: limit a range in which the video is acquired to a size of an image receiving unit for the radiation or to the irradiation range of the radiation; and control turning on and turning off the light based on the video acquired in the limited range.
- 7.** The information processing apparatus according to claim 1, wherein the processor is configured to control turning on and turning off the light based on a first mode in which the light is turned on and turned off using the acquired video or on a second mode in which the light is not turned on and turned off using the acquired video.
- 8.** The information processing apparatus according to claim 7, wherein the processor is configured to control turning on and turning off the light via the first mode in a state in which radiography is possible.
- 9.** The information processing apparatus according to claim 7, wherein the processor is configured to control turning on and turning off the light via the first mode in a case of radiography in a specific imaging menu.
- 10.** The information processing apparatus according to claim 7, wherein the processor is configured to control turning on and turning off the light via the first mode in a case of radiography at an upright position.
- 11.** The information processing apparatus according to claim 7, wherein the video acquired by the imaging apparatus is a visible light image.
- 12.** The information processing apparatus according to claim 7, wherein the video acquired by the imaging apparatus is a visible light image and a depth image corresponding to the visible light image.
- 13.** A non-transitory computer-readable storage medium storing an information processing program executable by a computer to execute a process comprising: acquiring a real-time video with an imaging apparatus; and controlling at least one of turning on or turning off light used to determine an irradiation range of radiation according to a movement of an object included in the acquired video.
- 14.** An information processing method executed by a computer, the method comprising: a step of acquiring a real-time video with an imaging apparatus; and a step of controlling at least one of turning on or turning off light used to determine an irradiation range of radiation according to a movement of an object included in the acquired video.
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